

Multiple Choice (10 marks)

1. What particle always has a mass of one atomic mass unit (amu) and no charge?
 - (a) a neutron
 - (b) a proton
 - (c) an electron
 - (d) an atom
2. Which sentence could not explain why the first place swimmer was faster than the second place swimmer?
 - (a) The winner experienced less water resistance.
 - (b) There was more gravitational pull on the winner.
 - (c) Friction slowed down the second place swimmer.
 - (d) There was more force in the winner's kicks and strokes.
3. The chemical symbol for niobium is Nb. Based on its symbol, what can be concluded about niobium?
 - (a) Niobium is a compound.
 - (b) Niobium is an element.
 - (c) Niobium is a metal.
 - (d) Niobium is a mixture.
4. How many times does Earth rotate on its axis in one day?
 - (a) once
 - (b) twice
 - (c) 24 times
 - (d) 365 times
5. Which organ in a frog has a function similar to the function of lungs in a bird?
 - (a) kidney
 - (b) skin
 - (c) liver
 - (d) heart

True / False (10 marks)

Answer True or False

6. True or False: The handle of a frying pan is often coated in rubber because rubber provides effective heat insulation.
7. True or False: A paramecium can conjugate, allowing it to exchange genetic material with another paramecium.
8. True or False: Both burning coal and exploding fireworks release energy during their chemical reactions.
9. True or False: The best way to organize plant growth data is by date, plant height, and soil type.
10. True or False: Rain can occur even when temperatures are below freezing, turning into snow upon contact with the ground.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the label of an object can change while its velocity remains constant, providing examples and analyzing the implications of such a scenario in motion.
12. Calculate the momentum of an 8.0 kg bowling ball moving at 2.0 m/s. Discuss the factors that influence momentum and its significance in real-world applications.

End of Paper